```
[root@DELL-RnD-India ~]# trace-cmd list | tail -10
nouserstacktrace
nosym-userobj
noprintk-msg-only
context-info
nolatency-format
sleep-time
graph-time
record-cmd
notest_nop_accept
notest_nop_refuse
[root@DELL-RnD-India ~]# trace-cmd record -e syscalls ls &> /dev/null
[root@DELL-RnD-India ~]# ls trace.dat
trace.dat
[root@DELL-RnD-India ~]# trace-cmd report | more
```

Figure 6: Using trace-cmd for recording and reporting

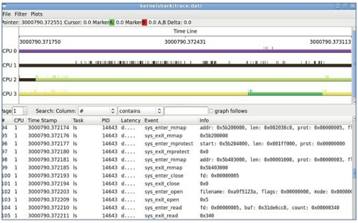

Figure 7: Analysing trace-cmd output with KernelShark

```
kvmmmu:kvm_mmu_sync_page
kvmmmu:kvm_mmu_unsync_page
kvmmmu:kvm_mmu_prepare_zap_page
kvm:kvm_entry
```

For example, to enable an event, you would use: *echo sys_enter_nice >> set_event* (note that you append the event name to the file, using the ">>" append redirector, and not ">"). To disable an event, precede the event name with a "!": *echo '!sys_enter_nice' >> set_event*. See Figure 5 for a sample event tracing scenario. The available events are listed in the events directory as well. For further details about event tracing, read the file *Documents/Trace/events.txt* in the kernel directory.

Trace-cmd and KernelShark

Trace-cmd, introduced by Steven Rostedt in his July 2009 post to the LKML (*http://marc.info/?l=linux-kernel&m=124784232716919&w=2*), makes it easy to manipulate the tracer. Follow these steps to get the latest version, including the GUI tool *KernelShark*, installed on your system:

```
wget http://www.hr.kernel.org/pub/linux/analysis/trace-cmd/trace-cmd-1.0.5.tar.gz
tar -zxvf trace-cmd-1.0.5.tar.gz
cd trace-cmd*
make
make gui # compiles GUI tools (KernelShark)
make install
make install_gui # installs GUI tools
```

With *trace-cmd*, tracing becomes a breeze (see Figure 6 for sample usage):

```
trace-cmd list  # to see available events
trace-cmd record -e syscalls ls # Initiate tracing on the syscall 'ls'
# (A file called trace.dat gets created in the current directory.)
trace-cmd report        # displays the report from trace.dat
```

KernelShark, installed by the *make install_gui* step above, can be used to analyse the trace data in the file *trace.dat*, as shown in Figure 7. END

References

You can find abundant information in the following resources, which I used as references for this article:

- The Linux kernel's Documentation/tracing directory
- Multiple articles from http://www.lwn.net
- KernelShark developer's page (http://rostedt.homelinux.com/kernelshark)

Acknowledgements

Special thanks to Srivathsa N S for helping me review this article

By: Surya Prabhakar

The author is an engineering advisor in the Product Group at Dell India R&D Centre, Bengaluru, and has eight years of experience in Linux. He spends most of his time hacking and playing around with Linux. He can be reached at surya_prabhakar@dell.com